

Heat stress Test on Agar

Overview

Applying heat stress to Arabidopsis on agar to evaluate which treatment time of CH bio peptide provides the best rescue effectiveness.

A. Materials

a. Agar

- i. $\frac{1}{2}$ MS medium powder
- ii. MES buffer
- iii. Distilled water
- iv. Sucrose
- v. Agar powder
- vi. pH meter
- vii. Flask
- viii. Stirring bar

b. Transfer

- i. Tweezer
- ii. New sets of neutral agar

c. Apply treatments

- i. CH Biotech peptide (富肽二號)
- ii. Sterile ddH₂O
- iii. 50ml centrifuge tube
- iv. Sprayer
- v. Incubator
- vi. Rack
- vii. Thermometer

B. Procedure

A. Neutral $\frac{1}{2}$ MS medium agar for vernalization

1. The following protocol is for 200 ml of agar
 - 4 neutral agar plates
2. Prepare the following:

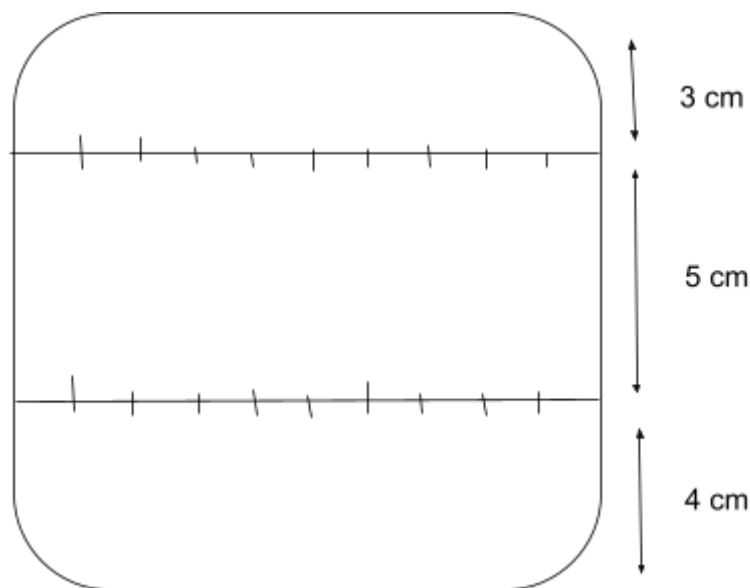
$\frac{1}{2}$ MS medium power	0.46g
MES (0.5g/L)	0.1g
Distilled water	200ml
Sucrose (1%)	2.0g

Agar powder (1%)	2.0g
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3. Add the contents into a 500ml flask
4. Use a magnetic stirrer to dissolve all ingredients (with a magnetic stirring rod inside) until no sediments are left in the solution
5. Use the pH meter and measure the pH of the solution, make sure the pH is 5.7-5.8 (Use KOH to adjust the medium pH)
6. Wrap the neck of the flask with aluminum foil and stick a piece of indicator tape on top
7. Autoclave at 127°C for 20 minutes
 - a. retrieving the flask until the pressure become 0 atm
8. During the autoclave, label all of the petri dishes with a marker in the biosafety cabinet
 - a. right top: date and arrow pointing up to show direction for pictures
9. Transfer half of the flasks to BSC
 - a. Store the other half of the flasks in the hot bath while pouring so that it does not solidify
10. Make sure the surface is dry by wiping the flasks with a paper towel before spraying 70% EtOH on them
11. Pour each of the medium mixtures into their square petri dishes (around 50ml/dishes)
12. Transfer the other half of the flasks into the BSC
13. Pour each of the medium mixtures into their square petri dishes
14. Let the petri dishes solidify at room temperature in the BSC

B. Planting and treating arabidopsis seeds

1. Open the BSC
2. Use the seeding template (3cm, 5cm, 4cm) to seeding 18 seeds on each agar plate in two rows, nine seeds on each row



3. Use breathable tape to seal the plates

4. Label the vernalization date
5. Wrap all agar plates with aluminum foil
6. Vernalize in the 4°C refrigerator for 4 days
7. Incubate in the growth chamber for 7 days to germination

C. Transferring seedling

1. Take the stack of agar plates wrapped in aluminum foil out of the refrigerator
2. Open the BSC
3. Spray alcohol on the agar plates before placing them inside the BSC
4. Use a marker to draw a straight line along a ruler 5.5cm from the top of the plate using the seed template
5. Place the square plates vertically at a slight angle
6. Choose 12 germinated seeds that have similar root lengths or germination status and align the endpoints of roots to the line drawn on the plate
 - a. 4 groups
 - i. Control
 - ii. Pre-treatment (1 day before)
 - iii. Co-treatment (day 0)
 - iv. After-treatment (1 day after)
7. Seal the plates with breathable tape and label the transferring date with a marker before placing the vertical plates on the rack in the growth chamber
 - a. Slightly tilted towards the back

D. Applying treatments

1. Spray CH bio peptide on pre-treatment group (1 day before)
 - a. Open the BSC
 - b. Dilute the CH bio peptide by 1/500 dilution rate (19.96ml of sterile ddH₂O + 40ul of peptide) into a 50ml centrifuge tube
 - c. Apply an individual sprayer onto the centrifuge tube
 - d. Take out the plate labeled “pre-treatment”, place it vertically, spray the CH bio peptide toward the plants entirely from leaves to roots with a 15cm distance in between (around 2 sprays, make sure all the plants received treatment)
 - e. Seal the plates with breathable tape and put it back into the growth chamber
2. Spray CH Biotech peptide(富肽二號) on co-treatment group(day 0)
 - a. Open BSC
 - b. Take out the plate labeled “co-treatment”, place it vertically, spray CH Biotech peptide toward the plants entirely with a 15cm distance in between (around 2 sprays, make sure all the plants are sprayed evenly)
 - c. Seal the plates with breathable tape

- d. Place all 4 groups of plates onto a rack while the plates stand with a slightly tilted angle, then place the rack entirely into the incubator at 37°C under white light for thermopriming
 - e. Observe the seedling for another 5 days in the incubator
 3. Spray CH bio peptide on after-treatment group(1 day after thermopriming)
 - a. Open BSC
 - b. Take out the plate labeled “after-treatment” from the incubator, place it vertically, spray CH bio peptide toward the plants entirely with a 15cm distance in between (around 2 sprays, make sure all the plants are sprayed evenly)
 - c. Seal the plates with breathable tape and place it back into the incubator

E. Recording (every day)

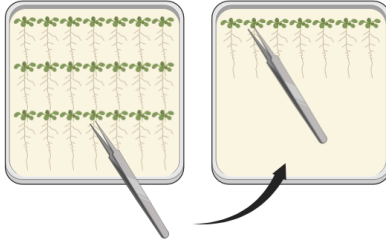
1. Observe for 6 days (days after pretreatment)
2. Set up the imaging platform with black background and white lights
3. Take out the agar plates one by one out of the growth chamber and place them on the platform
4. Take pictures (clearly record the labels, no reflections and avoid water droplet from covering the view of the plants)
5. Place the agar plates back to the growth chamber with tilted slightly to the back on the rack
6. After 6 days treatment, harvest the seedling and measure fresh weight, chlorophyll content for further analysis



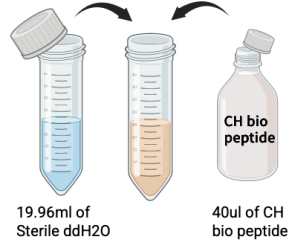
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(1 day before)

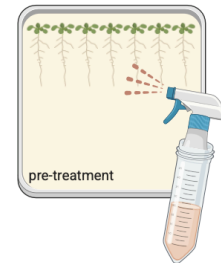
- ① Transfer 7 days old seedlings to a new neutral agar plate and label its corresponding group



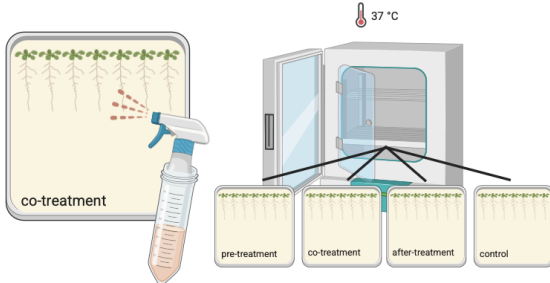
- ② Dilute the CH bio peptide by 1/500



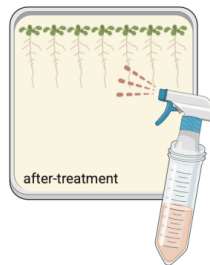
- ③ Spray diluted protectants to the **pre-treatment group** until all the leaves received treatment (around 2 sprays)



- (Day 0)
④ Spray diluted protectants to the **co-treatment group** and put all groups into the incubator for thermo-priming, observe for the next 5 days.



- (1 day after)
⑤ Spray diluted protectants to the **after-treatment group**



- ⑥ Take photos and record everyday

